

MODULE - I

Introduction to Cybercrime: Cybercrime: Definition and Origins of the Word, Cybercrime and Information Security, Who are Cybercriminals? Classifications of Cybercrimes, An Indian Perspective, Hacking and Indian Laws., Global Perspectives Textbook:1 Chapter 1 (1.1 to 1.5, 1.7-1.9)

- Countless number of websites that are unrestricted
- Internet has given way for Cyber crime
- The entities involved are Computers, internet, Cyberspace and World Wide Web (www)
- **Cyberspace:** It is where users mentally travel through matrices of data
 - Humans interact over the computer network
 - It is the world-wide network of computer networks that uses TCP/IP for communication to facilitate the exchange of data
- **Cybersquatting:** The term is derived from “squatting” which is the act of occupying an abandoned/unoccupied space/ building that the squatter does not own, rent or otherwise have permission to use. Cybersquatting is the act of registering a popular Internet address, usually a company name, with the intent of selling it to its rightful owner. From an affected individual's point of view, cybersquatting means register- ing, selling or using a domain name with the intent of profiting from the goodwill of someone else's trademark.
- **Cyberpunk and Cyberwarfare:** the words “cyber” and "punk" emphasize the two basic aspects of cyberpunk: “technology” and “individualism.” The term “cyberpunk” could mean some-thing like “anarchy via machines” or “machine/computer rebel movement.”. The idea behind calling it “cyber-punk” was to invent a new term that will express the juxtaposition of punk attitudes and high technology. For the terms “hackers,” “crackers”.

Cyberwarfare, for many people, means information warriors unleashing vicious attacks against an unsuspecting opponent's computer networks, wreaking havoc and paralyzing nations.

CYBER CRIME: DEFINITION AND ORIGIN OF THE WORD:

A crime conducted in which a computer was directly or significantly instrumental.

Definitions of Cyber Crime:

- Any illegal Act where a special knowledge of Computer Technology is essential for its Perception, Investigation or Prosecution.

- Any traditional crime that has acquired a new dimension through the aid of computer and abuses through computer.
- Any financial dishonesty that takes place in the computer environment
- Any threats to the computer such as HW/SW Threats, or Sabotage OR Demand for Ransom.
- Any illegal behaviour through electronic operations that targets the computer system and the data processed by them.

Specific Definitions:

- A crime committed using the computers and the internet to steal a person's identity or sell CONTRABAND or STALK or Disrupt operations through Victim's Malevolent programs.
- Crimes carried out either on computer or with computer
- Any illegal activity done through the internet or the computer
- All criminal activities done using the medium of computers, internet, cyberspace and www.

TWO TYPES OF ATTACK:**1. Techno Crime:**

An act against a system/systems with an intent to copy, steal, prevent access, corrupt or damage parts of a computer system. Can happen from anywhere in the world because of internet connectivity.

2. Techno-Vandalism:

Defacement of Websites such as copying or defacement of material or publicise the content.

Difference between “computer crime” and “computer fraud”:

There is a very thin line between the two terms “computer crime” and “computer fraud”; both are punishable Cybercrimes (harmful acts committed from or against a computer or network) differ from most terrestrial crimes in four ways: (a) how to commit them is easier to learn, (b) they require few resources relative to the potential damage caused, (c) they can be committed in a jurisdiction without being physically present in it and (d) they are often not clearly illegal.

Cyber fraud refers to the term where the users on the net are misguided and cheated and ultimately **fooling them**, whereas the **cyber crime refers to process of hacking**.

The term cybercrime has some stigma attached and is notorious due to the word “terrorism” or “terrorist” attached with it, that is, Cyberterrorism is defined as “any person, group or organization who, with terrorist intent, utilizes accesses or aids in accessing a computer or computer network or electronic system or electronic device by any available means, and thereby knowingly engages in or attempts to engage in a terrorist act commits the offence of cyberterrorism.”

CYBETCRIME AND INFORMATION SECURITY

-Lack of Information security leads to Cybercrime

CYBER SECURITY MEANS:

- Protecting information, equipment, devices, systems, communication devices and information stored from Unauthorised modifications/Deletions etc.

The term incorporates both the physical security of devices as well as the information stored therein. It covers protection from unauthorized access, use, disclosure, disruption, modification and destruction.

Cybercrimes occupy an important space in information security domain because of their impact.

For anyone trying to compile data on business impact of cybercrime, there are number of challenges.

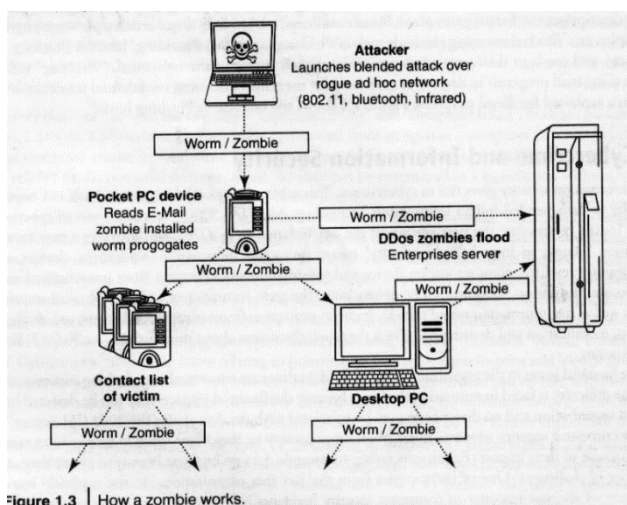
- One of them comes from the fact that organizations do not explicitly incorporate the cost of the vast majority of computer security incidents into their accounting as opposed to, say, accounting for the “shrinkage” of goods from retail stores.
- The other challenge comes from the difficulty in attaching a quantifiable monetary value to the corporate data and yet corporate data get stolen/lost

Because of these reasons, reporting of financial losses often remains approximate.

Types of network misuses are for:

- Internet Radio/Streaming Audio
- Streaming Video
- File Sharing
- Instant Messaging
- Online Gaming

to



BOTNET MENACE: The term “Botnet” is used refer to a group of compromised computers (zombie computers, i.e., personal computers secretly under the control of hackers) running malwares under a common command and control infrastructure. Figure 1.3 shows how a "zombie" works.

A Botnet maker can control the group remotely for illegal purposes, the most common

being denial-of-service attack (DoS attack), Adware, Spyware, E-Mail Spam, Click Fraud theft of application serial numbers, login IDs and financial information such as credit card numbers, etc. An attacker usually gains control by infecting the computers with a virus or other Malicious Code. The computer may continue to operate normally without the owner's knowledge that his computer has been compromised.

WHO ARE THE CYBER CRIMINALS?

What constitutes a Cyber Crime?

- Child Pornography
- Credit Fraud
- Cyber Stalking: is the use of the Internet or other electronic means to stalk or harass an individual, group, or organization.

It may include false accusations, defamation, slander(making a false statement damaging a person's reputation) and libel (posting or email causing dis-reputation). It may also include monitoring, identity theft, threats, vandalism, , doxing (*act of revealing identifying information about someone online*), or blackmail.

- Defaming another online
- Gaining unauthorised access to Computer System
 - Ignoring Copyright
 - Overriding encryption to make illegal copies.
 - Software Piracy and stealing another's identity (identity theft) to perform criminal acts.

TYPES OF CYBER CRIMINALS:

Type 1: Hungry for Recognition

- Hobby Hackers
- IT professionals
- Politically Motivated Hackers
- Terrorist Organizations

Type 2: Not interested in Recognition

- Psychological Perverts
- Financially motivated Hackers
- State Sponsored Hacking
- Organized Criminals

Type 3: The Insiders

- Disgruntled or former employees seeking revenge;
- Competing companies using employees to gain economic advantage through Damage.

CLASSIFICATION OF CYBER CRIMES:

	<i>Cybercrime in Narrow Sense</i>	<i>Cybercrime in Broad Sense</i>	
Role of computer	<i>Computer as an object</i> The computer/information stored on the computer is the subject/target of the crime	<i>Computer as a tool</i> The computer/or information stored on the computer constitutes an important tool for committing the crime	<i>Computer as the environment or context</i> The computer/information stored on the computer plays a non-substantial role in the act of crime, but does contain evidence of the crime
Examples	Hacking, computer sabotage, DDoS-attacks (distributed denial-of-service attacks), virtual child pornography	Computer fraud, forgery, distribution of child pornography	Murder using computer techniques, bank robbery and drugs trade

Crime is defined as “an act or the commission of an act that is forbidden, or the omission of a duty that is commanded by a public law and that makes the offender liable to punishment by that law” cybercrimes are classified as follows:

I. CYBER CRIME AGAINST INDIVIDUAL

1. **Electronic mail (e-mail) spoofing and other online frauds:** A spoofed E-Mail is one that appears to originate from one-source but has been sent from another source.

For example, let us say, Roopa has an E-Mail address roopa@asianlaws.org. someone spoofs her E-Mail and sends obscene/vulgar messages to all her acquaintances. Since the E-Mails appear to have originated from Roopa, her friends could take offense and relationships could be spoiled for life.

2. **phishing, spear phishing and its various other forms such as vishing and smishing:** Phishing (*the practice of sending fraudulent communications that appear to come from a reputable source*. It is usually done through email)

Spear Phishing (*a phishing method that targets specific individuals or groups within an organization*. It is a potent variant of phishing),

Smishing is a type of phishing scam where attackers send SMS messages (or text messages) to trick victims into sharing personal information or installing malware on their devices.

Vishing is almost identical to smishing, except cybercriminals use VoIP (Voice over IP) to place phone calls to trick victims instead of SMS (short message service) messages.

3. **Spamming:** - Spam is the abuse of electronic messaging systems (including most broadcast media, digital delivery systems) to send unsolicited bulk messages indiscriminately. Although the most widely recognized form of Spam is E-Mail Spam, the term is applied to similar abuses in other media: instant messaging Spam, Usenet newsgroup Spam, web search engine Spam, Spam in blogs, wiki Spam, online classified ads Spam, mobile phone messaging Spam, Internet forum Spam, junk fax transmissions, social networking Spam, file sharing network Spam, video sharing sites, etc.

Spamming is difficult to control because it has economic viability —

- advertisers have no operating costs beyond the management of their mailing lists, and
- it is difficult to hold senders accountable for their mass mailings.
- Spammers are numerous; the volume of unsolicited mail has become very high because: the barrier to entry is low.

Spamming has economic implications:

- Results in Loss of Productivity and fraud borne by public
- ISPs have to add extra capacity to cope with this.

Spamming Definition 2:

- Alteration or creation of a document to deceive a filing system.

Some web authors use “subversive techniques” to ensure that their site appears more frequently or higher number in returned search results

A FEW TIPS: Avoid the following web publishing techniques

- Repeating Key Words
- Use of Key words that do not related to the site content
- Use of Fast Meta Refresh (The *meta-refresh tag*, or meta redirect, is one way that you can reload or redirect web pages)
- Redirection
- IP Cloaking (is when a person hides their IP address.)
- Use of coloured text on the same colour background.
- Duplication of pages with same URLs
- Hidden Links
- Use of different pages that bridge to the same URL

4. **Cyberdefamation:** -Defamation means, Make or publish any imputation for a person intending to harm the reputation of that person.

Cyber Defamation: Electronic Form of defamation. According to IPC Section 499 defamation is said to happen in the following cases

- To impute anything to a deceased person or hurt the reputation of a living person or to his family.
- To make imputation against a company or a group of persons
- Expression Ironically
- Lowers the Moral and intellectual character of that person.

LIBEL: Written Defamation

SLANDER: Oral Defamation

5. **Cyberstalking and Harassment:** the dictionary meaning of "stalking" is an "act or process of following prey stealthily - trying to approach somebody or something." cyberstalking has been defined as the use of information and communications technology, particularly the internet, by an individual or group of individuals to harass another individual, group of individuals, or organization.

The behaviour includes false accusations, monitoring, transmission of threats, id theft, damage to data or equipment, solicitation of minors for sexual purposes, and gathering information for harassment purposes.

6. **Computer Sabotage:** The use of the internet to hinder the normal functioning of a computer system through the introduction of worms, viruses, or logic bombs, is referred to as computer sabotage.

It can be used to gain economic advantage over a competitor, to promote the illegal activities of terrorists or to steal data or programs for extortion purposes.

- logic bombs are event-dependent programs created to do something only when a certain event (known as a trigger event) occurs.
- some viruses may be termed as logic bombs because they lie dormant all through the year and become active only on a particular date.

7. **Pornographic Offenses:** Child Pornography: Any photo or video that can be considered obscene and /or unsuitable for the age of a child. The Internet is being highly used by its abusers to reach and abuse children sexually, worldwide

Here is how the 'Pedophiles' act:

Step 1: Use a false identity to trap the child/teenager

Step 2: They get into a play area where the children are playing

Step 3: They befriend children and gain confidence

Step 4: They extract personal information

Step 5: Get email address and make contact

Step 6: They start sending pornographic images/textual messages to the child. They create an impression that it is a normal act

Step 7: Finally, they set up a face to face meeting and sexually assault the child.

8. **Password Sniffing:** Password sniffers are programs that monitor and record the name and password of network users as they login, jeopardizing security at a site. whoever installs the sniffer can then impersonate an authorized user and login to access restricted documents.

II. CYBERCRIME AGAINST PROPERTY

1. **Credit Card Frauds:** Online Scams, There are a few major types of crimes under the category of hacking: Spoofing website and E-Mail security alerts, hoax mails about virus threats, lottery frauds and Spoofing.
 - In Spoofing websites and E-Mail security threats, fraudsters create authentic looking websites that are nothing but a spoof .

The purpose of these websites is to make the user enter personal information which is then used to access business and bank accounts. Fraudsters are increasingly turning to E-Mail to generate traffic to these websites. This kind of online fraud is common in banking and financial sector.

- It is wise to be alert and careful about E-Mails containing an embedded link, with a request for you to enter secret details. It is strongly recommended not to input any sensitive information that might help criminals to gain access to sensitive information, such as bank account details, even if the page appears legitimate.
- In virus hoax E-Mails, the warnings may be genuine, so there is always a dilemma whether to take them lightly or seriously. A wise action is to first confirm by visiting an antivirus site such as McAfee, Sophos or Symantec before taking any action, such as forwarding them to friends and colleagues.
- Lottery frauds are typically letters or E-Mails that inform the recipient that he/she has won a prize in a lottery. To get the money, the recipient has to reply, after which another mail is received asking for bank details so that the money can be directly transferred.
- “Spoofing” means illegal intrusion, posing as a genuine user. A hacker logs-in to a computer illegally, using a different identity than his own. He is able to do this by having previously obtained the actual password. He creates a new identity by fooling the computer into thinking that the hacker is the

genuine system operator and then hacker then takes control of the system. He can commit innumerable number of frauds using this false identity,

2. **Intellectual Property Crimes:** Software Piracy, Copyright Infringement, Trademark Violations, Theft of Computer Source Code etc.
3. **Internet Time Theft:** Such a theft occurs when an unauthorized person uses the Internet hours paid for by another person.

Basically, Internet time theft comes under hacking because the person who gets access to someone else's ISP user ID and password, either by hacking or by gaining access to it by illegal means, uses it to access the Internet without the other person's knowledge. The issue of Internet time theft is related to the crimes conducted through "identity theft."

III. CYBER CRIME AGAINST ORGANIZATION:

1. **Unauthorised accessing of Computers:** Although the purposes of hacking are many, the main ones are as follows:
 - a) Greed;
 - b) power;
 - c) publicity;
 - d) revenge;
 - e) adventure;
 - f) desire to access forbidden information;
 - g) destructive mindset.

Every act committed toward breaking into a computer and/or network is hacking and it is an offense. Hackers write or use ready-made computer programs to attack the target computer. They possess the desire to destruct and they get enjoyment out of such destruction.

Some hackers hack for personal monetary gains, such as stealing credit card information, transferring money from various bank accounts to their own account followed by withdrawal of money. Government websites are hot on hackers' target lists and attacks on Government websites receive wide press coverage

Hackers, crackers and phrackers! are some of the oft-heard terms, The meaning has now changed to become something associated with the breaking into or harming of any kind of computer or telecommunications system. Some people claim that those who break into computer systems should ideally be called “crackers” and those targeting phones should be known as “phreaks”

2. **Password Sniffing:** Password Sniffers are programs that monitor and record the name and password of network users as they login, jeopardizing security at a site. Whoever installs the Sniffer can then impersonate an authorized user and login to access restricted documents
3. **Denial of Service (DoS) attacks:**A denial-of-service attack (DoS attack) is an attempt to make a computer resource (i.e., information systems) unavailable to its intended users.

In this type of criminal act, the attacker

- floods the bandwidth of the victim's network or
- fills his E-Mail box with Spam mail depriving him of the services he is entitled to access or provide.

Although the means to carry out, motives for, and targets of a DoS attack may vary, it generally consists of the concerted efforts of a person or people to prevent the Internet site or service from functioning efficiently or at all, temporarily or indefinitely.

Symptoms of DoS attacks to include:

- Unusually slow network performance (opening files or accessing websites);
- Unavailability of a particular website;
- Inability to access any website;
- Dramatic increase in the number of Spam E-Mails received (this type of DoS attack is termed as an E-Mail bomb).

The goal of DoS is not to gain unauthorized access to systems or data, but to prevent intended users (i.e., legitimate users) of a service from using it.

A DoS attack may do the following:

- Flood a network with traffic, thereby preventing legitimate network traffic.
- Disrupt connections between two systems, thereby preventing access to a service.
- Prevent a particular individual from accessing

4. **Virus Attack/Dissemination:** Computer virus is a program that can "infect" legitimate programs by modifying them to include a possibly "evolved" copy of itself.

Viruses spread themselves, without the knowledge or permission of the users, to potentially large numbers of programs on many machines. A computer virus passes from computer to computer in a similar manner as a biological virus passes from person to person. Viruses may also contain malicious instructions that may cause damage or annoyance; the combination of possibly Malicious Code with the ability to spread is what makes viruses a considerable concern. Viruses can often spread without any readily visible symptoms. A virus can start on event-driven effects (e.g., triggered after a specific number of executions), time-driven effects (e.g., triggered on a specific date, such as Friday the 13th) or can occur at random. Viruses can take some typical actions:

- Display a message to prompt an action which may set off the virus; delete files inside the system into which viruses enter;
 - Scramble data on a hard disk;
 - Cause erratic screen behaviour;
 - Halt the system (PC);
 - Just replicate themselves to propagate further harm.
5. **E-Mail bombing/mail bombs:** E-Mail bombing refers to sending a large number of E-Mails to the victim to crash victim's E-Mail account (in the case of an individual) or to make victim's mail servers crash (in the case of a company or an E-Mail service provider).

Computer program can be written to instruct a computer to do such tasks on a repeated basis.

In recent times, terrorism has hit the Internet in the form of mail bombings. By instructing a computer to repeatedly send E-Mail to a specified person's E-Mail address, the cybercriminal can overwhelm the recipient's personal account and potentially shut down entire systems.

6. **Salami attack/Salami technique:** These attacks are used for committing financial crimes.

The idea here is to make the alteration so insignificant that in a single case it would go completely unnoticed; for example, a bank employee inserts a program, into the bank's servers, that deducts a small amount of money (say Rs.2/- or a few cents in a month) from the account of every customer.

No account holder will probably notice this unauthorized debit, but the bank employee will make a sizable amount every month, Small "Shavings" for Big Gains!.

7. **Logic bomb: (Computer Sabotage).** The use of the internet to hinder the normal functioning of a computer system through the introduction of worms, viruses, or logic bombs, is referred to as computer sabotage.

It can be used to gain economic advantage over a competitor, to promote the illegal activities of terrorists or to steal data or programs for extortion purposes.

- logic bombs are event-dependent programs created to do something only when a certain event (known as a trigger event) occurs.
- some viruses may be termed as logic bombs because they lie dormant all through the year and become active only on a particular date.

8. **Trojan Horse:** Trojan Horse is a program in which malicious or harmful code is contained inside apparently harmless programming or data in such a way that it can get control and cause harm, for example, ruining the file allocation table on the hard disk.

Unlike viruses or worms, Trojans do not replicate themselves but they can be equally destructive. On the surface, Trojans appear benign and harmless, but once the infected code is executed, Trojans kick in and perform malicious functions to harm the computer system without the user's knowledge.

9. **Data diddling:** (data is altered as it is entered into a computer system, most often by a data entry clerk or a computer virus. Computerized processing of the altered data results in a fraudulent benefit. In some cases, the altered data is changed back after processing to conceal the activity. The results can be huge. They might include adjusting financial figures up or down marginally, or it could be more complex and make an entire system unusable. Electricity Boards in India have been victims to data diddling programs inserted when private parties computerize their systems.

10. **Crimes emanating from Usenet newsgroup:** This is one form of spamming. The word "Spam" was usually taken to mean excessive multiple posting (EMP). The advent of Google Groups, and its large Usenet archive, has made Usenet more attractive to spammers than ever. Spamming of Usenet newsgroups actually predates E-Mail Spam. The first widely recognized Usenet Spam titled Global Alert for All: Jesus is Coming Soon (though not the most famous) was posted on 18 January 1994.

11. **Industrial spying industrial espionage:** Spying is not limited to governments. Corporations, like governments, often spy on the enemy. The Internet and privately networked systems provide new and better opportunities for espionage. "Spies" can get information about product finances, research and development and marketing strategies, an activity known as "industrial spying."

Highly skilled hackers, contracted by high-profile companies or certain governments via the means of

escrow organizations (it is said that they get several hundreds of thousands of dollars, depending on the “assignment”).

Organizations subject to online extortion tend to keep quiet about it to avoid negative publicity about them.

12. **Computer network intrusions**: Computer Networks pose a problem by way of security threat because people can get into them from anywhere.

The popular movie “War Games” illustrated an extreme but useful example of this. “Crackers” who are often misnamed “Hackers”! can break into computer systems from anywhere in the world and steal data, plant viruses, create backdoors, insert Trojan Horses or change user names and passwords. Network intrusions are illegal, but detection and enforcement are difficult. Current laws are limited and many intrusions go undetected.

13. **Software piracy**: “software piracy” as theft of software through the illegal copying of genuine programs or the counterfeiting and distribution of products intended to pass for the original. There are many examples of software piracy:

- end-user copying ~ friends loaning disks to each other, or organizations under-reporting the number of software installations they have made, or organizations not tracking their software licenses;
- hard disk loading with illicit means ~ hard disk vendors load pirated software;
- counterfeiting — large-scale duplication and distribution of illegally copied software;
- illegal downloads from the Internet — by intrusion, by cracking serial numbers, etc.

Beware that those who buy pirated software have a lot to lose:

- (a) getting untested software that may have been copied thousands of times over,
- (b) the software, if pirated, may potentially contain hard-drive-infecting viruses,
- (c) there is no technical support in the case of software failure, that is, lack of technical product support available to properly licensed users,
- (d) there is no warranty protection, (e) there is no legal right to use the product, etc.

IV. CYBER CRIME AGAINST SOCIETY

1. **Forgery**: Counterfeit currency notes, postage and revenue stamps, marksheets, etc. can be forged using sophisticated computers, printers and scanners. Outside many colleges there are miscreants soliciting the sale of fake marksheets or even degree certificates, these are made using computers and high-quality scanners and printers. In fact, this is becoming a booming business involving large monetary amount given to student gangs in exchange for these bogus but authentic looking certificates.

2. **Cyber Terrorism:** Cyberterrorism is defined as “any person, group or organization who, with terrorist intent, utilizes accesses or aids in accessing a computer or computer network or electronic system or electronic device by any available means, and thereby knowingly engages in or attempts to engage in a terrorist act commits the offence of cyberterrorism.” Cybercrime, especially through the Internet, has grown in number as the use of computer has become central to commerce, entertainment, and government.
3. **Web Jacking:** Web jacking occurs when someone forcefully takes control of a website (by cracking the password and later changing it). Thus, the first stage of this crime involves “password sniffing.” The actual owner of the website does not have any more control over what appears on that website.

V. CRIMES EMINATING FROM USENET NEWSGROUP

By its very nature, Usenet groups may carry very offensive, harmful, inaccurate or otherwise inappropriate material, or in some cases, postings that have been mislabelled or are deceptive in another way. Therefore, it is expected to use caution and common sense and exercise proper judgment when using Usenet, as well as use the service at one's own risk

CYBERCRIMES: INDIAN PERSPECTIVE

- 4th Highest number of Internet users
- increase in Cybercrime from 2006-2007.
- 46% of the total cases related to Cyberpornography followed by Hacking
-

Special Cybercrime cell in Delhi:

- Over 100 Police Officers regularly trained in tackling Cybercrime.
- Domains of Training: [Computer Networking, Data Communication, Protocols, Wireless Networks etc]

Cybercrime Indian Statistics

(A) Cybercrimes: Cases of Various Categories under ITA 2000

217 cases were registered under IT Act during the year 2007 as compared to 142 cases during the previous year (2006), India is said to be the "youth country" given the population age distribution. From the potential resources perspective, this is supposed to be a great advantage; assuming that these youths will get appropriate training to develop the required professional skills in them. However, from cybercrime perspective, this youth aspect does not seem good as revealed by cybercrime statistics in India. Crime head-wise and age-group-wise profile of the offenders arrested under ITA 2000 revealed that 55.8% (86 out of 154) of the offenders were arrested under "Obscene publication/ transmission in electronic form" of which

70.9% (61 out of 86) were in the age group]8-30 years.

50% (24 out of 48) of the total persons arrested for "Hacking with Computer Systems" were in the age group of 18-30 years.

(B) Cybercrimes: Cases of Various Categories under IPC Section

A total of 339 cases were registered under IPC Sections during the year 2007 as compared to 311 such cases during 2004, thereby reporting an increase of 9.0%.

(C) Incidence of Cybercrimes in Cities

17 out of 35 mega cities did not report any case of cybercrime (neither under the IT Act nor under IPC Sections) during the year 2007. A total of 17 mega cities have reported 118 cases under IT Act and 7 mega cities reported 180 cases under various sections of IPC. There was an increase of 32.6% (from 89 cases in 2006 to 118 cases in 2007).

CYBERCRIME AND THE INDIAN ITA 2000

In India, the ITA 2000 was enacted after the United Nation General Assembly Resolution A/RES/51/162 in January 30, 1997 by adopting the Model Law on Electronic Commerce adopted by the United Nations Commission on International Trade Law. This was the first step toward the Law relating to E-Commerce at international level to regulate an alternative form of commerce and to give legal status in the area of E-commerce.

It was enacted taking into consideration UNICITRAL model of Law on Electronic Commerce (1996).

Hacking and the Indian Laws:

Cybercrimes are punishable under two categories:

(i) ITA 2000

(ii) Indian Penal Code (IPC)

Table 1.7 | The key provisions under the Indian ITA 2000 (before the amendment)

<i>Section Ref. and Title</i>	<i>Chapter of the Act and Title</i>	<i>Crime</i>	<i>Punishment</i>
Sec. 43 (Penalty for damage to computer, computer system, etc.)	CHAPTER IX Penalties and Adjudication	Damage to computer system, etc.	Compensation for ₹ 1 crore (₹ 10,000,000).
Sec. 66 (Hacking with computer system)	CHAPTER XI Offences	Hacking (with intent or knowledge).	Fine of ₹ 2 lakhs (₹ 200,000) and imprisonment for 3 years.
Sec. 67 (Publishing of information which is obscene in electronic form)	CHAPTER XI Offences	Publication of obscene material in electronic form.	Fine of ₹ 1 lakh (₹ 100,000), imprisonment of 5 years and double conviction on second offence.

<i>Section Ref. and Title</i>	<i>Chapter of the Act and Title</i>	<i>Crime</i>	<i>Punishment</i>
Sec. 68 (Power of controller to give directions)	CHAPTER XI Offences	Not complying with directions of controller.	Fine up to ₹ 2 lakhs (₹ 200,000) and imprisonment of 3 years.
Sec. 70 (Protected system)	CHAPTER XI Offences	Attempting or securing access to computer of another person without his/her knowledge.	Imprisonment up to 10 years.
Sec. 72 (Penalty for breach of confidentiality and privacy)	CHAPTER XI Offences	Attempting or securing access to computer for breaking confidentiality of the information of computer.	Fine up to ₹ 1 lakh (₹ 100,000) and imprisonment up to 2 years.
Sec. 73 (Penalty for publishing Digital Signature Certificate false in certain particulars)	CHAPTER XI Offences	Publishing false digital signatures, false in certain particulars.	Fine of ₹ 1 lakh (₹ 100,000) or imprisonment of 2 years or both.
Sec. 74 (Publication for fraudulent purpose)	CHAPTER XI Offences	Publication of Digital Signatures for fraudulent purpose.	Imprisonment for the term of 2 years and fine of ₹ 1 lakh (₹ 100,000).

GLOBAL PERSPECTIVES ON CYBERCRIME: At an international level, there is a broad scope and coverage for handling Cybercrimes. In Europe, there is a treaty called “Council of Europe (COE). - Cybercrime is an umbrella. The offences include:

- Computer Data and Systems
- Content Offences
- Copyright Offences
- White Collar crimes not only involving Computers.

About 30 countries have enacted an Anti-Spam legislation.

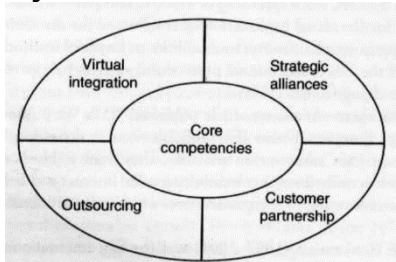
Spamming is increasingly on the rise:

- Financial offenses including attempts to capture from “So called Trusted” Companies
- Vehicle to spread Viruses
- Since the crime can originate from any country across the borders and the Victims could be anywhere in the interconnected world, there is a strong need for international cooperation to solve the global cybercrimes.
- Linking of Cybersecurity with Critical Infrastructure protection is very much needed

A Few developments with Global Cybersecurity:

1. **August 4, 2006 Announcement:** US senate ratifies COE convention on Cybercrime. It targets Hackers, People spreading disruptive computer viruses, using internet for sexual exploitation etc.
2. **August 18, 2001:** There was a news article on “ISPS wary about Drastic Oblocations on Website Blocking”. EU wished to debar suspicious websites.
 - Has the power to block website that incite terrorist activities.
3. **COE Cybercrime Convention (1997-2000):**
 - First international treaty seeking to address internet crimes, improve investigative techniques and increase cooperation between nations.

Cybercrime and the Extended Enterprise:



The term (Fig.below) represents the concept that a company is made up not just of its employees, its board members and executives, but also its business partners, its suppliers and even its customers.

The extended enterprise can only be successful if all of the component groups and individuals have the information, they need in order to do business effectively. An extended enterprise is a “loosely coupled, self-organizing network” of firms that combine their economic output to provide “products and services” offerings to the market.

Firms in the extended enterprise may operate independently, for example, through market mechanisms or cooperatively through agreements and contracts.

Seamless flow of “information” to support instantaneous “decision-making ability” is crucial for the “external enterprise.” This becomes possible through the “interconnectedness.” Due to the interconnected features of information and communication technologies, security overall can only be fully promoted when the users have full awareness of the existing threats and dangers. Governments, businesses and the international community must, therefore, proactively help users access information on how to protect themselves.

Given the promises and challenges in the extended enterprise scenario, organizations in the international community have a special role in sharing information on good practices, and creating open and accessible enterprise information flow channels for exchanging of ideas in a collaborative manner, International cooperation at the levels of government, industry, consumer, business and technical groups to allow a global and coordinated approach to achieving global cybersecurity is the key.